Movement of a metal ball in a wire field — Solution

X23 (2023) — English translation

A1 **0.80 pts**

A metal ball of radius R is placed in a uniform electric field $\vec{E}_0$. Find the dipole moment $\vec{p}$ acquired by the ball. Express your answer in terms of $\vec{E}_0$, R and the electric constant ϵ_0.

The electric field inside the ball is zero, and outside the ball it is the sum of the external field and the dipole field:

$$\vec{E} = \vec{E}_0 + \frac{1}{4\pi\epsilon_0}\left(\frac{3\left(\vec{p}\cdot\vec{r}\right)\vec{r}}{r^5} - \frac{\vec{p}}{r^3}\right)$$

On the surface of the ball, the electric field is directed perpendicular to the surface. From here:

Answer:

$$\vec{p} = 4\pi\epsilon_0 R^3 \vec{E}_0$$

A2 **0.50 pts**

Let the ball be located in an electric field directed along the z axis, the intensity of which varies according to the law:

$$\vec{E} \approx \left(E_0 + kz\right)\vec{e}_z$$

where E_0 and k — known quantity, moreover $kR \ll E_0$. Find force F_z, act on sphere. Answer express through E_0, k, R and electric constant ϵ_0.

By definition, a dipole is two point charges $+q$ and $-q$ located at a distance l from each other. Let $\vec{r}$ be the radius vector of the negative charge, and $\vec{l}$ be the radius vector of the positive charge relative to the negative one. Then for the interaction force we have:

$$\vec{F} = q\left(\vec{E}(\vec{r}+\vec{l}) - \vec{E}(\vec{r})\right)$$

If electric field direct along the dipole moment system, that, since dipole is oriented along axis z, force, act on it, direct along axis z. If dipole one can consider/count point

$$E_z(\vec{r}+\vec{l}) - E_z(\vec{r}) = \frac{dE_z}{dz}\cdot l$$

Hence for force obtain:

$$F_z = p_z\frac{dE_z}{dz}$$

in our problem since $kR \ll E_0$ - model electric dipole well describes electric field sphere. Substitute expression for p_z and $\frac{dE_z}{dz}$, we find:

Answer:

$$F_z = 4\pi R^3\epsilon_0 E_0 k$$

A3 **0.20 pts**

Find the magnitude of the electric field of the wire E_w at a distance r from it. Express your answer in terms of λ, r and the electric constant ε_0.

From Gauss's theorem we have:

$$2\pi r L E_{\mathrm{w}} = \frac{q}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$

whence:

Answer:

$$E_{\mathrm{w}} = \frac{\lambda}{2\pi\epsilon_0 r}$$

A4 — 0.50 pts

Find the magnitude and direction (attraction or repulsion) of the force F acting on the ball from the side of the wire at a distance r from it. Express your answer in terms of λ, R, r and the electric constant ε_0.

At $r \gg R$ the values E and $\frac{dE}{dr}$ of the wire fields on the ball size scale can be considered constant. In this approximation, the dipole moment of the ball is equal to:

$$p = \frac{2\lambda R^3}{r}$$

Force, act on sphere, equal:

$$F_r = p\frac{dE_r}{dx}$$

whence:

Answer: The force of attraction is:

$$F = \frac{\lambda^2 R^3}{\pi\epsilon_0 r^3}$$

A5 — 0.40 pts

Show that the mechanical potential energy W_p of a ball in the field of a wire at a distance r between the wire and the center of the ball is equal to:

$$W_p = -\frac{A}{r^2}$$

Find A. Answer express through λ, R and electric constant ε_0.

For potential energy we have:

$$W_p = -\int_r^{\infty} F dr$$

Integrate, find:

$$W_p = -\frac{\lambda^2 R^3}{2\pi\epsilon_0 r^2}$$

Hence parameter A is equal to:

Answer:

$$A = \frac{\lambda^2 R^3}{2\pi\epsilon_0}$$

B1 — 0.60 pts

Find the minimum distance $r_{\min}$ between the center of the ball and the wire during motion, assuming that no collision occurs. Express your answer in terms of A, b, m and v_0.

From the law of conservation of energy we have:

$$\frac{mv^2}{2} - \frac{A}{r^2} = \frac{mv_0^2}{2}$$

We can represent square velocity in following form:

$$v^2 = \dot{r}^2 + r^2\dot{\varphi}^2$$

From the law conservation angular momentum obtain:

$$L = mr^2\dot{\varphi} \Rightarrow \dot{\varphi} = \frac{v_0 b}{r^2}$$

Combining with law conservation energy, obtain:

$$\frac{m\dot{r}^2}{2} + \frac{mv_0^2 b^2}{2r^2} - \frac{A}{r^2} = \frac{mv_0^2}{2}$$

Minimum distance to wire be achieved for $\dot{r} = 0$. From here we get:

Answer:

$$r_{min} = \sqrt{b^2 - \frac{2A}{mv_0^2}}$$

B2 — 0.20 pts

Find v_{crit}. Express your answer in terms of A, b and m.

The radical expression must be non-negative. From here we get:

Answer:

$$v_{crit} = \frac{1}{b}\sqrt{\frac{2A}{m}}$$

C1 — 0.40 pts

Express u' in terms of r, $\dot{r}$ and $\dot{\varphi}$. Find $u'(0)$ corresponding to the value of $\varphi = 0$.

Explicitly differentiate u:

$$u' = \frac{du}{d\varphi} = -\frac{1}{r^2}\frac{dr}{d\varphi}$$

We can represent $\frac{dr}{d\varphi}$ as derivative complex function:

$$\frac{dr}{d\varphi} = \frac{\dot{r}}{\dot{\varphi}}$$

from where:

The value is constant and equal to:

$$r^2\dot{\varphi} = \frac{L}{m} = v_0 b$$

At the initial moment, the speed is directed directly towards the wire. Then we find:

Answer:

$$u' = -\frac{\dot{r}}{r^2\dot{\varphi}}$$

C2 — 0.30 pts

Express the mechanical energy E of the ball in terms of m, u, u', v_0, b and A.

The expression for the mechanical energy of the ball is as follows:

$$E = \frac{m\dot{r}^2}{2} + \frac{mr^2\dot{\varphi}^2}{2} - \frac{A}{r^2}$$

From angular momentum conservation have:

$$L = mr^2\dot{\varphi} = mv_0 b$$

from which:

$$E = \frac{m\dot{r}^2}{2} + \frac{mv_0^2 b^2}{2r^2} - \frac{A}{r^2}$$

Transition to variable u and u', we obtain:

Answer:

$$\frac{mv_0^2 b^2}{2}\left(u'^2 + u^2\right) - Au^2 = E = \frac{mv_0^2}{2}$$

C3 0.70 pts

Get addicted $u(\varphi)$. Express your answer in terms of m, v_0, b, A and φ.

Differentiating the expression for energy with respect to φ, we obtain the equation of harmonic vibrations:

$$u'' = -u\left(1 - \frac{2A}{mv_0^2 b^2}\right)$$

with cyclic frequency:

$$\omega_0 = \sqrt{1 - \frac{2A}{mv_0^2 b^2}}$$

General solution write in form:

$$u(\varphi) = C\sin\left(\omega_0\varphi + \varphi_1\right)$$

Taking into account initial condition:

$$u(0) = 0 \quad u'(0) = \frac{1}{b}$$

We find:

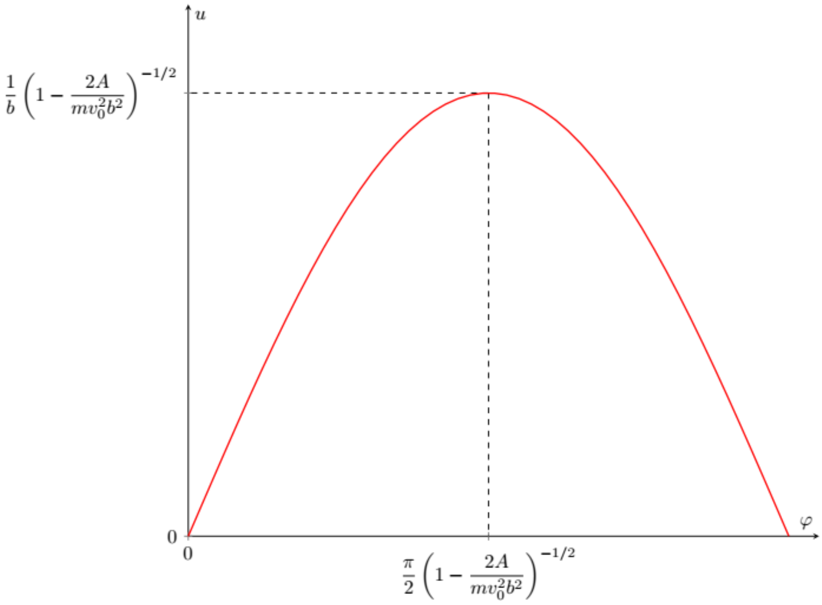
Answer:

$$u(\varphi) = \frac{1}{b\sqrt{1 - \frac{2A}{mv_0^2 b^2}}}\sin\left(\varphi\sqrt{1 - \frac{2A}{mv_0^2 b^2}}\right)$$

C4 0.60 pts

On the answer sheet, qualitatively construct the trajectory of the center of the ball in coordinates u,φ. The coordinates of all feature points can be expressed in terms of m, v_0, b and A.

The value u, starting from zero, reaches a maximum, and then, reaching zero, remains constant in time. The graph is shown in the figure below:



C5 — 0.50 pts

Using the value of the parameter A found in step **A4**, *find the initial speed of the ball v_1 at which during the interaction between it and the wire the direction of the ball's speed changes to the opposite. Express your answer in terms of m, b, R, λ and the electrical constant ϵ_0.*

When the direction of the ball's speed is reversed, the total change in angle φ is 2π. From here we find the value ω_0:

$$\omega_0 = \frac{1}{2}$$

Then parameter A equal:

$$A = \frac{3mv_1^2b^2}{8}$$

Equate with expression, obtain in part **A4**

$$\frac{\lambda^2R^3}{2\pi\epsilon_0} = \frac{3mv_1^2b^2}{8}$$

we find:

Answer:

$$v_1 = \frac{2\lambda R}{b}\sqrt{\frac{R}{3\pi\epsilon_0 m}}$$

D1 — 0.60 pts

Find S and q_2. Express your answers using R, L and q.

Let us introduce a coordinate system at the center of a cylindrical sphere with the axis z directed along the line connecting the center of the sphere with the charge q_1. The potential of charges q_1 and $-q_2$ at any point on the surface of the ball is equal to zero, therefore:

$$\frac{q_1}{\sqrt{r^2+(x-L)^2}} = \frac{q_2}{\sqrt{r^2+(x-S)^2}}$$

or:

$$(q_1^2-q_2^2)(r^2+x^2)+2x(q_2^2L-q_1^2S)+q_1^2S^2-q_2^2L^2=0$$

Since $r^2 + x^2 = R^2$, coefficient for x equal zero. Then obtain the system of equations:

$$\frac{L}{S} = \frac{q_1^2}{q_2^2} \qquad (q_1^2 - q_2^2)R^2 = q_2^2 L^2 - q_1^2 S^2$$

solving which, we find:

Answer:

$$S = \frac{R^2}{L} \quad q_2 = \frac{q_1 R}{L}$$

D2

0.40 pts

Find dq_1, dq_2, S and r_-. Express your answers using λ, R, r, α and $d\alpha$.

For dq_1 we have:

$$dq_1 = \lambda r d \tan\alpha$$

or:

The element in question is located at a distance $L = \dfrac{r}{\cos\alpha}$ from the center of the ball. From here we find dq_2 and S:

For r_- we have:

$$r_- = r - S\cos\alpha$$

or

Answer:

$$dq_1 = \frac{\lambda r}{\cos^2\alpha} d\alpha$$

D3

0.20 pts

Find the magnitude and direction (attraction or repulsion) of the vector sum of forces $d\vec{F} = d\vec{F}_+ + d\vec{F}_-$ interaction of charges dq_2 and $-dq_2$ with the wire. Express your answer in terms of λ, R, r, α, $d\alpha$ and the electrical constant ϵ_0.

The pair of image charges are attracted to the wire because the negative charge is at a smaller distance from the wire. For the force of attraction, taking into account the electric field of the wire, we have:

$$dF = \frac{\lambda dq_2}{2\pi\epsilon_0}\left(\frac{1}{r_-} - \frac{1}{r}\right)$$

or:

Answer: The force of attraction is

$$dF = \frac{\lambda^2 R^3}{2\pi\epsilon_0 r^3} \frac{\cos\alpha d\alpha}{1 - \dfrac{R^2}{r^2} + \dfrac{R^2}{r^2}\sin^2\alpha}$$

D4

1.00 pts

Find the magnitude and direction of the total force of interaction between the ball and the wire F_{tot}. Express your answer in terms of λ, R, θ and the electric constant ϵ_0.

To find the total force, we calculate the integral of the expression obtained in point **D3** in the range from $-\dfrac{\pi}{2}$ to $\dfrac{\pi}{2}$. Let's introduce the value t, equal to:

$$t = \frac{R\sin\alpha}{\sqrt{r^2 - R^2}}$$

Then obtain:

$$F = \frac{\lambda^2 R^2}{2\pi\epsilon_0 r\sqrt{r^2 - R^2}} \int\limits_{-\frac{R}{\sqrt{r^2-R^2}}}^{\frac{R}{\sqrt{r^2-R^2}}} \frac{dt}{1+t^2} = \frac{\lambda^2}{\pi\epsilon_0} \frac{R}{r} \frac{R}{\sqrt{r^2 - R^2}} \arctan \frac{R}{\sqrt{r^2 - R^2}}$$

Account for relation r, R and θ, we find:

Answer: The force of attraction is

$$F = \frac{\lambda^2}{\pi\epsilon_0} \frac{\sin^2\theta \cdot \theta}{\cos\theta}$$

D5 1.00 pts

Find the potential energy W_p of the interaction of the ball with the wire. Express your answer in terms of λ, R, θ and the electric constant ε_0.

Considering that the ball is attracted to the wire, the expression for potential energy is:

$$W_p = -\int\limits_r^{\infty} F(r) dr$$

Find quantity dr in variable R and θ:

$$dr = d\left(\frac{R}{\sin\theta}\right) = -\frac{R\cos\theta d\theta}{\sin^2\theta}$$

Then have:

$$W_p = -\frac{\lambda^2 R}{\pi\epsilon_0} \int\limits_0^{\theta} \theta d\theta$$

from where:

Answer:

$$W_p(\theta) = -\frac{\lambda^2 R \theta^2}{2\pi\epsilon_0}$$

D6 0.30 pts

What speed v_{st} will an uncharged metal ball of radius R and mass m have when colliding with a wire if it is released from infinity without an initial speed? Express your answer in terms of λ, R, m and the electric constant ε_0.

At the moment of contact between the ball and the wire $\theta = \frac{\pi}{2}$. Let's write down the law of conservation of energy:

$$\frac{mv_0^2}{2} = \frac{\lambda^2 R}{2\pi\epsilon_0}\left(\frac{\pi}{2}\right)^2$$

from where:

Answer:

$$v = \frac{\pi\lambda}{2}\sqrt{\frac{R}{\pi\epsilon_0 m}}$$

E1 — 0.40 pts

For a given value v_0, obtain the dependence of the square of the radial velocity component $\dot{r}^2$ on the parameter θ. Express your answer in terms of m, v_0, λ, R, b, electric constant ε_0 and θ.

From the law of conservation of energy we have:

$$\frac{mv^2}{2} - \frac{\lambda^2 R\theta^2}{2\pi\epsilon_0} = \frac{mv_0^2}{2}$$

We can represent velocity in following form:

$$v^2 = \dot{r}^2 + r^2\dot{\varphi}^2$$

Taking into account law conservation moment momentum

$$L = mv_0 b = mr^2\dot{\varphi}$$

we find:

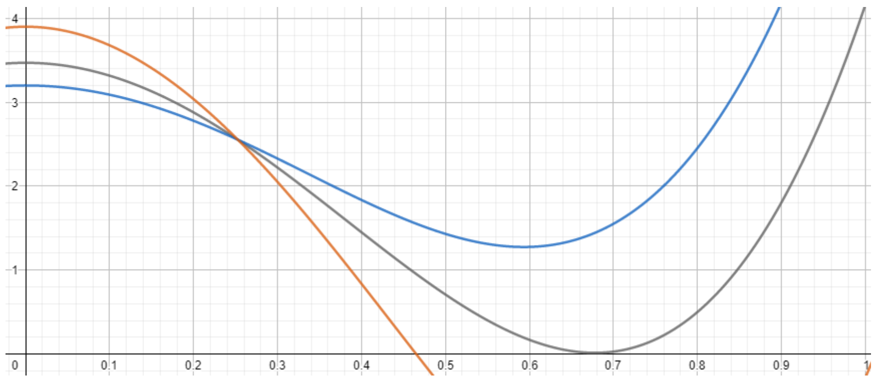
Answer:

$$\dot{r}^2 = v_0^2 - v_0^2\frac{b^2}{R^2}\sin^2\theta + \frac{\lambda^2 R}{\pi\epsilon_0 m}\theta^2$$

E2 — 0.60 pts

Construct high-quality graphs of the dependence $\dot{r}^2(\theta)$ for different values of v_0.

Depending on the value of v_0, functions are divided into three types: 1) intersecting the ordinate axis at $\theta < \frac{\pi}{2}$; 2) ordinates that do not intersect the axis; 3) touching the ordinate axis at $\theta < \frac{\pi}{2}$.



E3 — 0.60 pts

Considering all quantities except v_0 constant, you obtain a system of equations that allows you to determine the value v_{crit}. The system of equations can include m, R, b, λ, electric constant ε_0, θ_{max} and v_{crit}.

At sufficiently high speeds (type 1 function), the graph intersects the ordinate axis. At the moment of intersection, the value of $\dot{r}$ is reversed because $\dot{r}^2 > 0$. At sufficiently low speeds (type 2 function), the value of $\dot{r}^2$ is positive at the extremum, so the ball falls on the wire. The function of the 3rd type is a boundary case of those considered above - at the extremum, the value of the function $\dot{r}^2$ becomes zero. At any speed, the ball will not touch the wire. From here it is clear that when the value $\dot{r}^2$ is equal to zero, the derivative $\frac{d\dot{r}^2}{d\varphi}$ must also is equal to to zero, which is written in the form of a system as follows:

Answer:

$$\left(\frac{b^2\sin^2\theta_{max}}{R^2} - 1\right) = \frac{\lambda^2 R\theta_{max}^2}{mv_0^2\pi\epsilon_0} \qquad \frac{b^2\sin 2\theta_{max}}{R^2} = \frac{2\lambda^2 R\theta_{max}}{mv_0^2\pi\epsilon_0}$$

E4 **1.20 pts**

Find v_{crit1}, v_{crit2} and v_{crit3} corresponding to the following values of b :

$$b_1 = 20R \quad b_2 = 3R \quad b_3 = 1{,}1R$$

Answer express through m, R, λ and electric constant ε_0.

From the equation for the zero derivative we find:

$$v_0 = \lambda\sqrt{\frac{R}{\pi\epsilon_0 m}} \cdot \frac{R}{b}\sqrt{\frac{2\theta_{max}}{\sin 2\theta_{max}}}$$

For finding v_0 necessarily determine θ_{max}. Divide equation one with another, obtain:

$$\frac{k^2\sin^2\theta_{max} - 1}{k^2\sin 2\theta_{max}} = \frac{\theta_{max}}{2} \quad k = \frac{b}{R}$$

Solve equation on the calculator, find:

$$\theta_{max1} = 0{,}296 \text{ rad} \quad \theta_{max2} = 0{,}793 \text{ rad} \quad \theta_{max3} = 1{,}459 \text{ rad}$$

from where:

Answer:

$$v_{crit1} \approx 5{,}15 \cdot 10^{-2}\lambda\sqrt{\frac{R}{\pi\epsilon_0 m}} \quad v_{crit2} \approx 3{,}15 \cdot 10^{-1}\lambda\sqrt{\frac{R}{\pi\epsilon_0 m}} \quad v_{crit3} \approx 3{,}30\lambda\sqrt{\frac{R}{\pi\epsilon_0 m}}$$